Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To Eliminate the concern of Diesel leak in Marazzo model"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	01.12.2018	05.12.2018
Completion Date	30.04.2019	12.05 .2019

Current Status: **Closed**

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarize & Plan future actions
- Guidelines to user

Cross Functional Team

No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Nilesh Kalekar	L7 - O	CAI	9850222441
2	**Mentor**	Rahul Narkar	HOD	STCF	9922409136
3	**Mentor**	Mandar Joshi	L6-M	MQA	8378991344
4	**Member 2**	Mangesh Musale	L7 - O	MQA	9730163917
5	**Member 3**	Pushpendra D	L7 - M	R&D	9823147429
6	**Member 4**	Kaka Sawant	L7 - M	PET	8237028924
7	**Member 5**	Gokul Nikam	L6-M	PVT	9923282427

Step1. Select Subject

Brief Description :

WHAT	Diesel leak
WHERE	In Field , Plot 126 and CAI squeak & rattle test.
WHEN	After running
WHO	Customer , By Squeak & rattle inspector
WHY	1. V1 Defect 2. CAI STR Loss 3. High Rework time 4. Customer complaint
HOW	Avg 20% of Marazzo vehicles

Diesel leak is repeat V1+ severity defect in field & CAI since Sept-18.

2. Justify the choice (reasons)

PROMISE 2022
AUTOMOTIVE SECTOR

We Will Grow

Revenue 75% | Profit 100%

By

'Enhancing lifestyle' in personal segment
'Enriching livelihood' in commercial segment

Being #1 in UV, SCV, EV and #3 in HCV
Doubling volume from overseas markets

Leading in next generation mobility solutions

#InspiringPeopleToRise

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice

Field Complaint

SCR No. S-1000000237773

SCR Status Submitted

Customer Complaint Details

Customer Complaint Details	FUEL LEAKAGE DUE TO BROKEN FUEL RETURN LINE.
Break-Down	Yes
Observations & Analysis	Vehicle reported with complain of fuel leakage as per customer verbatim he observed fuel smell during driving thus to check he stopped the vehicle & found fuel was leaking inside engine compartment. Thus he bought vehicle to workshop. In inspection we found fuel (Diesel) was leaking from Injector fuel return line we found fuel return line hose in broken condition due to excess bent point in return hose.
Action Taken	New part wasn't available thus we have cut the hose & refitted it after proper rerouting of hose to avoid future failure which was caused due to incorrect routing. Relevant photos, videos attached for better reference.

ip	1053317.00 KB
zip	11.00 KB
	0 KB
	0 KB
g1.zip	14.00 KB

Customer reacted on social media

SCR Details

SCR No.	S-1000000237773
SCR Submitted Date	28-Dec-2018
Serial Number	J2L10044
Engine Number	ESJZK*****
ModelCode	AU32BPES7T151A00WP
Description	MAHINDRA MARAZZO 7 STR M2 WHT
ModelFamily	765
Description	MARAZZO
Customer Name	MAHINDRA & MAHINDRA LTD
Mileage	6055
Vehicle Registration No.	J2L10044
City Of Operation	GHAZIABAD
Month & Year Of Sale	-
Date Of Failure	28/12/2018 00:00:00
Type Of Complaint	Un-scheduled
Verbatim Group Code	ENG-[Engine]
Verbatim Code	ENG1030-Leak - Fuel Lines

2. Justify the choice

Imp. Problem Selection:-

- Diesel leak is V1+ severity concern – if defective vehicle goes to customer / LSP inspector / CQA Auditor's who performs dynamic test of vehicle.

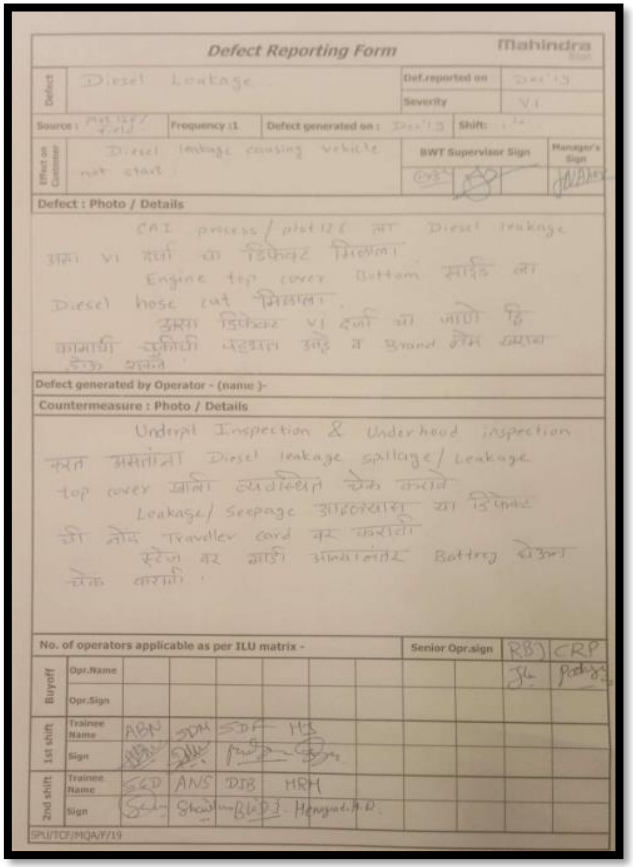
- It is also a customer irritant if goes directly to the customer.

Step2.b Immediate Containment Action

Customer Protection –

- Since this problem being detected at Squeak & rattle stage so the customer is protected.

DRF to be given to stage associate & Repair stage associate for Vehicle.



Defect Reporting Form

Mahindra

Defect: Diesel Leakage

Def.reported on: Dec'13

Severity: V1

Source: Mat. Ins / Field

Frequency: 1

Defect generated on: Dec'13

Shift:

Effect on Customer: Diesel leakage causing vehicle not start

BWT Supervisor Sign

Manager's Sign

Defect : Photo / Details

CAI process / plot 12 E का Diesel leakage आया V1 दर्जा का डिफेक्ट मिलता। Engine top cover Bottom साईड का Diesel hose cut मिलता। इसका डिफेक्ट V1 दर्जा का जाता है कामायनी कुमियों नहीं आई व Brand ठीक नहीं हो सकता।

Defect generated by Operator - (name)-

Countermeasure : Photo / Details

Underpit Inspection & Underhood inspection करते समय Diesel leakage spillage / Leakage top cover अलग अलग चेक करना Leakage / Seepage आहलयास या डिफेक्ट की नोंद Traveller card पर कराना इंजन पर गाड़ी आने के बाद Battery कनेक्शन चेक करना।

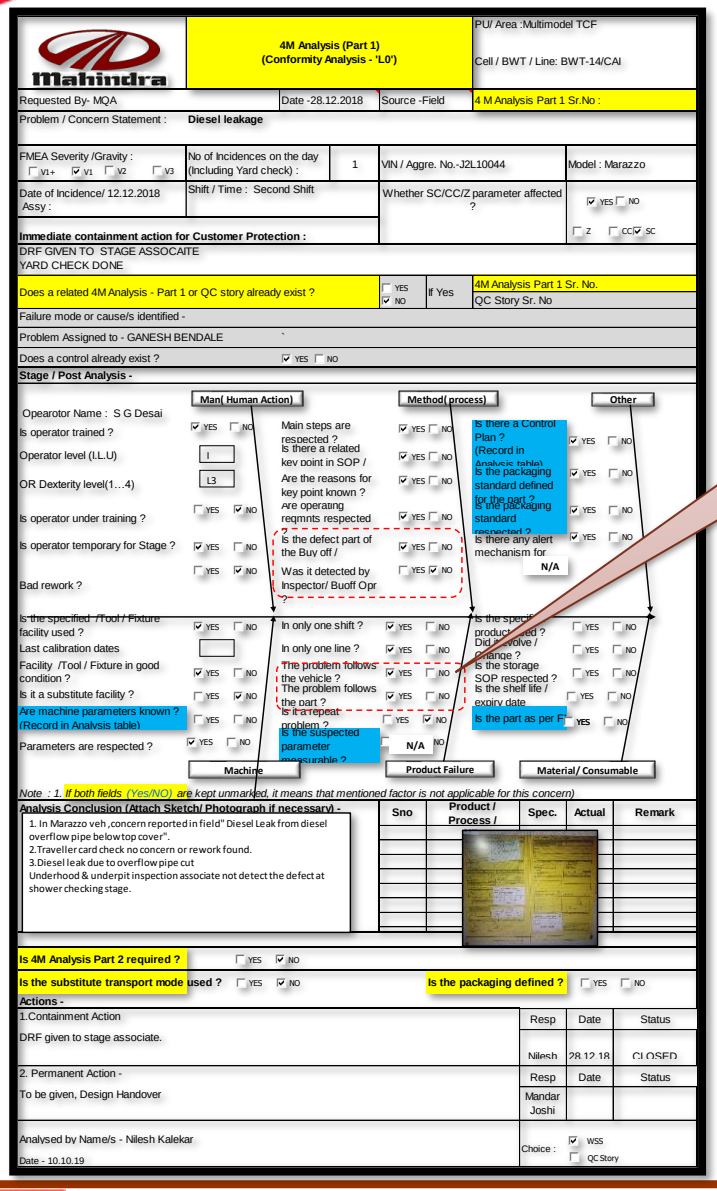
No. of operators applicable as per ILU matrix -

Senior Opr.sign: RB | CRP

Buyoff	Opr.Name										
	Opr.Sign										
1st shift	Trainee Name	ABN	SDM	SDF	MS						
	Sign										
2nd shift	Trainee Name	SGD	ANS	DIB	MRH						
	Sign										

SPU/TCF/MQA/F/19

3. Understand the current situation



Mahindra

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area :Multimodel TCF
Cell / BWT / Line: BWT-14/CAI

Requested By- MQA | Date -28.12.2018 | Source -Field | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : **Diesel leakage**

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.-J2L10044 | Model : Marazzo

Date of Incidence/ 12.12.2018 Assy : | Shift / Time : Second Shift | Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☐ CC ☑ SC

Immediate containment action for Customer Protection :
DRF GIVEN TO STAGE ASSOCAITE
YARD CHECK DONE

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. / QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - GANESH BENDALE

Does a control already exist ? ☑ YES ☐ NO

Stage / Post Analysis -

Man(Human Action) | Method(process) | Other

Opearotor Name : S G Desai
Is operator trained ? ☑ YES ☐ NO
Operator level (I.L.U) I
OR Dexterity level(1…4) L3
Is operator under training ? ☐ YES ☑ NO
Is operator temporary for Stage ? ☑ YES ☐ NO
Bad rework ? ☐ YES ☑ NO

Main steps are respected ? ☑ YES ☐ NO
Is there a related key point in SOP / ☑ YES ☐ NO
Are the reasons for key point known ? ☑ YES ☐ NO
Are operating reqmnts respected ☑ YES ☐ NO
Is the defect part of the Buy off / ☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ? ☐ YES ☑ NO

Is there a Control Plan ? (Record in Analysis table) ☑ YES ☐ NO
Is the packaging standard defined for the part ? ☑ YES ☐ NO
Is the packaging standard respected ? ☑ YES ☐ NO
Is there any alert mechanism for N/A ☑ YES ☐ NO

Is the specified /Tool / Fixture facility used ? ☑ YES ☐ NO
Last calibration dates
Facility /Tool / Fixture in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☐ YES ☑ NO
Are machine parameters known ? (Record in Analysis table) ☐ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO

In only one shift ? ☑ YES ☐ NO
In only one line ? ☑ YES ☐ NO
The problem follows the vehicle ? ☑ YES ☐ NO
The problem follows the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☐ YES ☑ NO
Is the suspected parameter measurable ? N/A ☐ NO

Is the specified product used ? Did it evolve / change ? ☐ YES ☐ NO
Is the storage SOP respected ? ☐ YES ☐ NO
Is the shelf life / expiry date ☐ YES ☐ NO
Is the part as per F YES ☐ NO

Machine | Product Failure | Material/ Consumable

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
1. In Marazzo veh ,concern reported in field" Diesel Leak from diesel overflow pipe below top cover".
2.Traveller card check no concern or rework found.
3.Diesel leak due to overflow pipe cut Underhood & underpit inspection associate not detect the defect at shower checking stage.

Sno	Product / Process /	Spec.	Actual	Remark

Is 4M Analysis Part 2 required ? ☐ YES ☑ NO
Is the substitute transport mode used ? ☐ YES ☑ NO **Is the packaging defined ?** ☐ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
DRF given to stage associate.	Nilesh	28.12.18	CLOSED

2. Permanent Action -	Resp	Date	Status
To be given, Design Handover	Mandar Joshi		

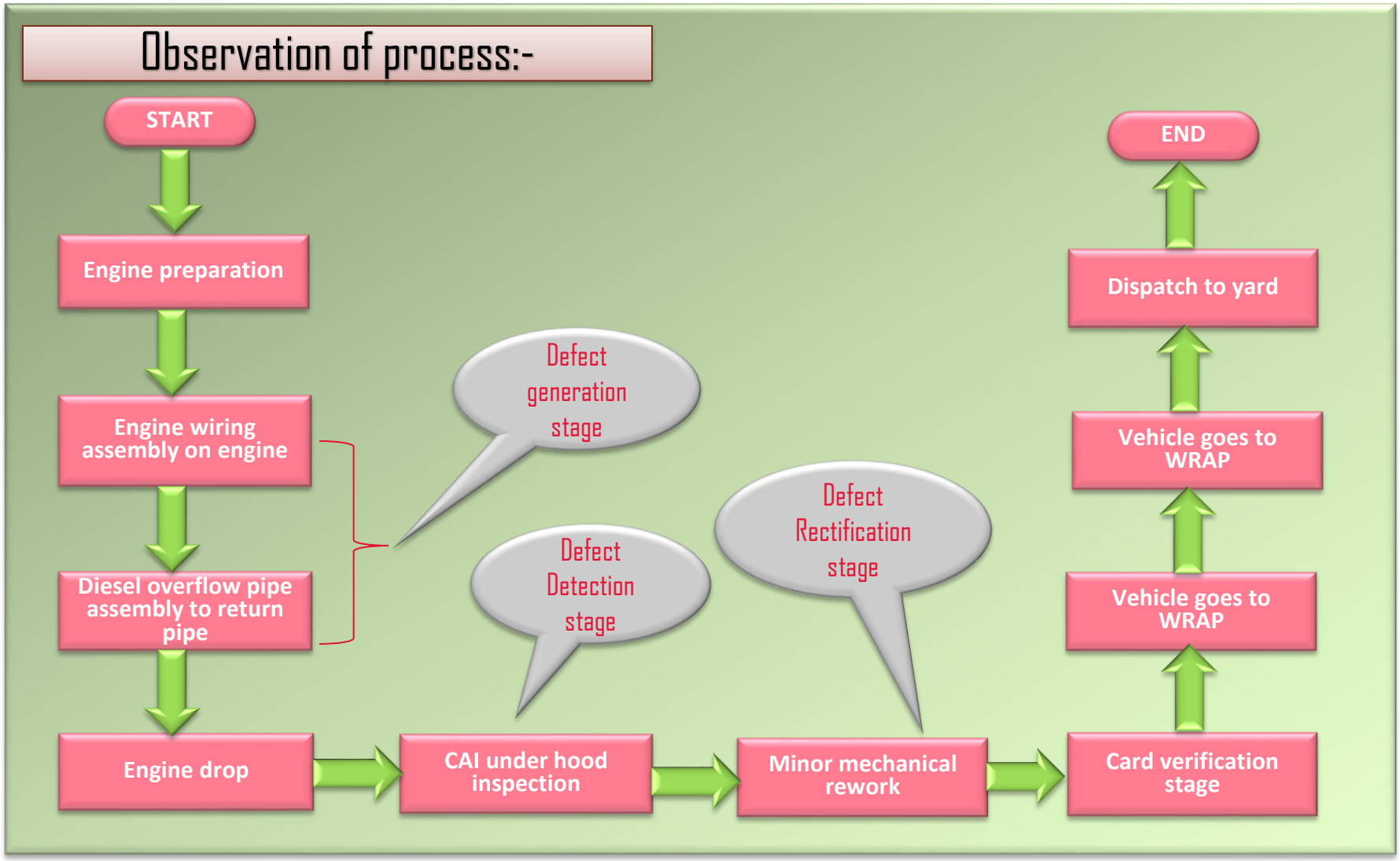
Analysed by Name/s - Nilesh Kalekar
Date - 10.10.19
Choice : ☑ WSS ☐ QC Story

Inference from L0,
Problem follows part & vehicle
Problem is a repeat problem &
hence detailed analysis required

No Quick win

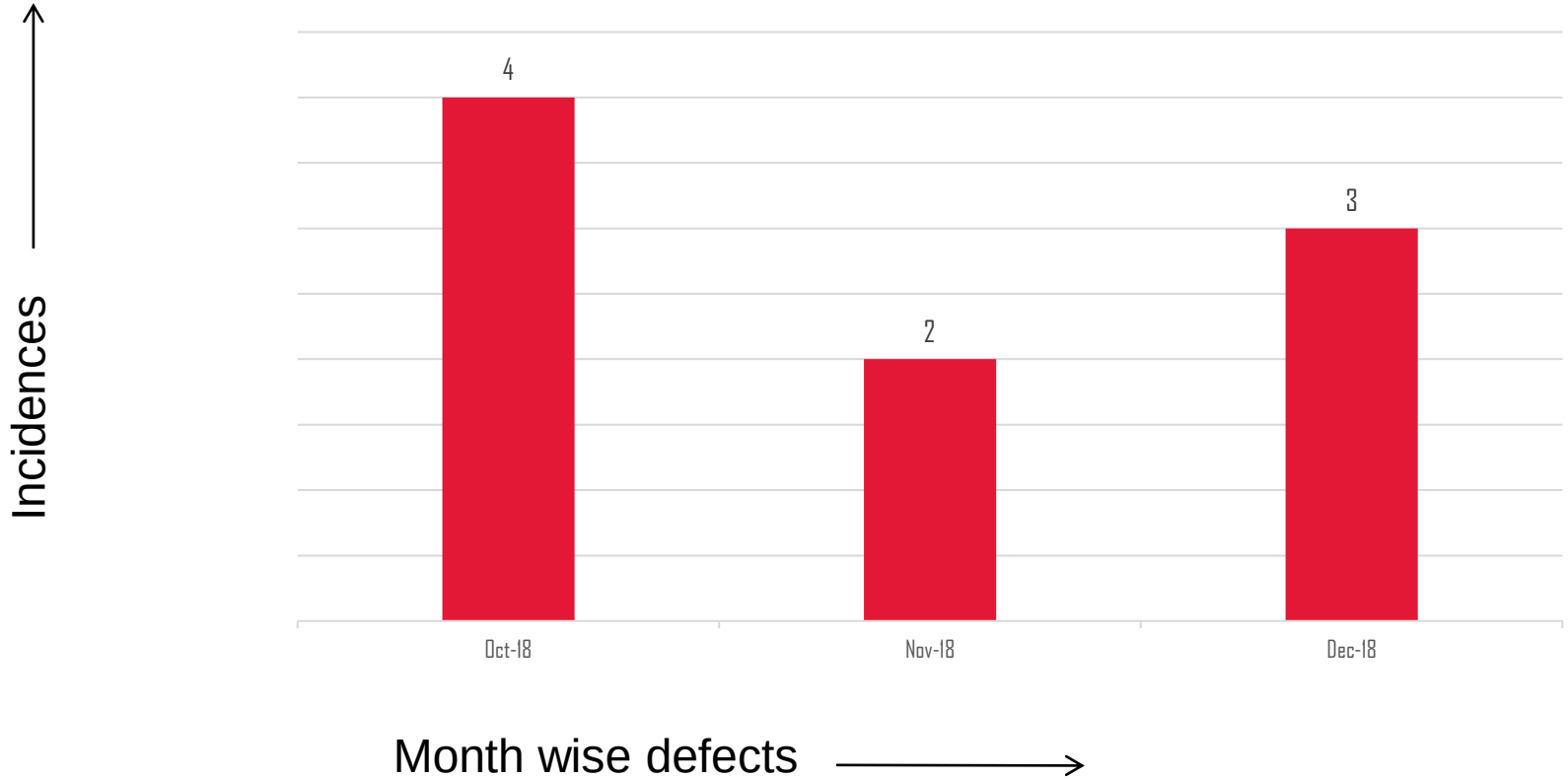
4M analysis Good and Not so good observations	
MAN	Operator Dext. Level L4
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows with vehicle
Material	Handling and shelf life of part adhered

Step 3. Understanding Current Situation



Step3. Understand the current situation

Month wise Trend For Diesel Leak at CAI & Field



Inference:-
From above data the diesel leak defect comes at mechanical repair stage at an average of 80 incidences per month.

Step 3. Understanding Current Situation

Good observations:

- Associate follows SOP for the wire routing, overflow pipe routing etc.
- Associate certifies the all fitment by self certification marking on parts.
- Defect closed at minor or major mechanical stage.
- Issue reported in both shifts.
- Not all the vehicles having this issue.

Not so Good observations:

- Sharp bend at corner
- Vehicle reported in CAI, at plot 126 & in field

4. Select targets

Present Level:

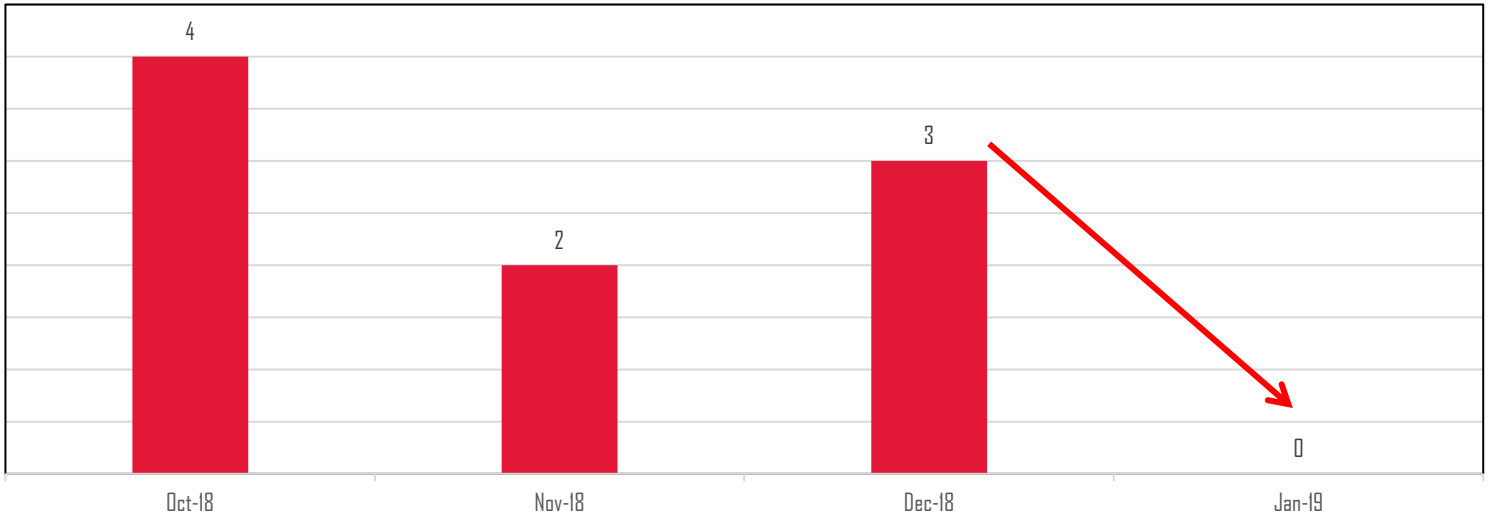
3 incidences reported for diesel leak in field ,

2 incidences reported for diesel leak at plot 126 &

4 incidence reported for diesel leak in CAI by squeak & rattle test inspector from past 3 months data

Target:

Elimination of Concern by Jan-2019



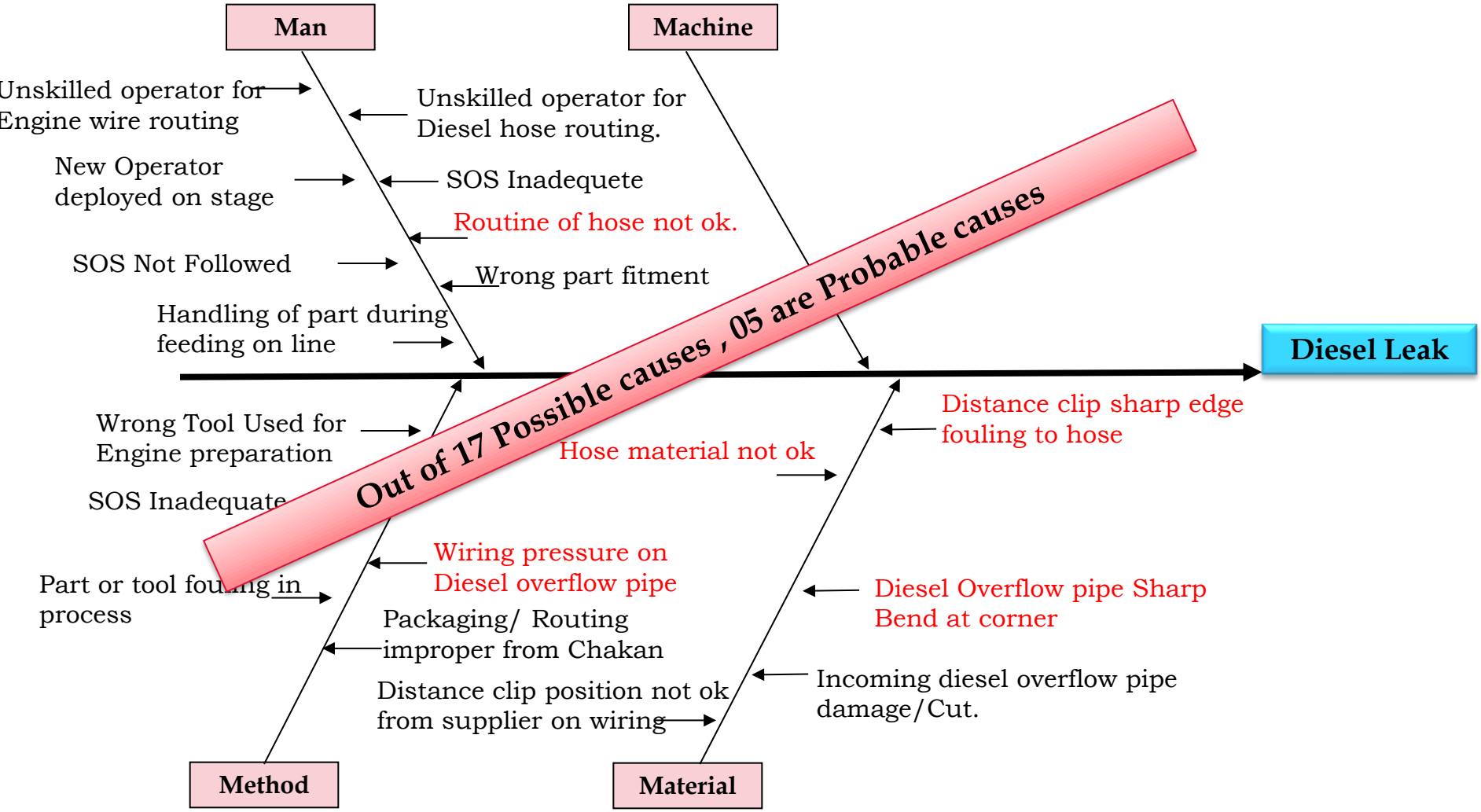
Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr. no	**Possible Causes**
1	New operator deployed at stage.
2	SOS Not Followed .
3	Unskilled operator for Engine wire routing
4	Unskilled operator for Diesel hose routing
5	Incoming diesel pipe damage/Cut.
6	Diesel Overflow pipe Sharp Bend at corner
7	Distance clip sharp edge fouling to hose
8	Distance clip position not ok from supplier on wiring
9	Wiring pressure on Diesel overflow hose pipe
10	Hose material not ok
11	Wrong Tool Used for Engine preparation
12	Part or tool fouling in process
13	Routine of hose not ok.
14	Wrong part fitment
15	SOS Inadequete
16	Packaging/ Routing improper from Chakan
17	Handling of part during feeding on line

Step 5. Analysis

List of possible causes Cause & Effect Diagram



Step 5. Analysis

5.I List of possible causes identification in Brain storming session

Sr. no	Possible Causes	Nilesh	Mandar Joshi	Mangesh Musale	Pushpendra	Kaka sawant	Gokul Nikam	Satish Pagar	Total
1	New operator deployed at stage.	1	1	3	3	3	3	3	17
2	SOS Not Followed .	1	1	1	1	1	1	1	7
3	Unskilled operator for Engine wire routing	3	3	3	3	3	9	9	33
4	Unskilled operator for Diesel hose routing	3	1	3	1	3	3	3	17
5	Incoming diesel pipe damage/Cut.	1	1	1	1	1	1	1	7
6	Diesel Overflow pipe Sharp Bend at corner	9	9	9	3	3	9	3	45
7	Distance clip sharp edge fouling to Diesel overflow Hose	1	3	9	9	9	3	9	43
8	Distance clip position not ok from supplier on wiring	1	1	3	3	1	3	3	15
9	Wiring pressure on Hose	9	9	3	3	9	3	3	39
10	Hose material not ok	9	9	9	3	3	3	9	45
11	Wrong Tool Used for Engine preparation	1	1	1	1	1	1	1	7
12	Part or tool fouling in process	1	1	1	1	1	1	1	7
13	Routine of hose not ok.	9	9	9	3	3	3	9	45
14	Wrong part fitment	1	1	1	1	1	1	1	7
15	SOS Inadequete	1	1	3	3	3	3	3	17
16	Packaging/ Routing improper from Chakan	3	1	3	1	3	1	3	15
17	Hanldling of part during feeding on line	1	1	1	3	1	1	3	11

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 39.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr.No	Not considered Causes	Reasons
1	New operator deployed at stage.	All trained operator deployed on stage.
2	SOS Not Followed .	SOS Followed 15 cycle observed.
3	Unskilled operator for Engine wire routing	All trained operator deployed on stage
4	Unskilled operator for Diesel hose routing	All trained operator deployed on stage
5	Incoming diesel pipe damage/Cut.	Visually checked confirmed OK.
8	Distance clip position not ok from supplier on wiring	20 wiring checked & confirmed vendor process.
11	Wrong Tool Used for Engine preparation	Tool used as per control plan no chance of fouling
12	Part or tool fouling in process	Process walk done no fouling to that area
14	Wrong part fitment	No other similar looking part at Chakan
15	SOS Inadequete	Checked & found ok
16	Packaging/ Routing improper from Chakan	Checked & found ok as per design routine
17	Hanldling of part during feeding on line	Checked & found ok

Step 5. Analysis

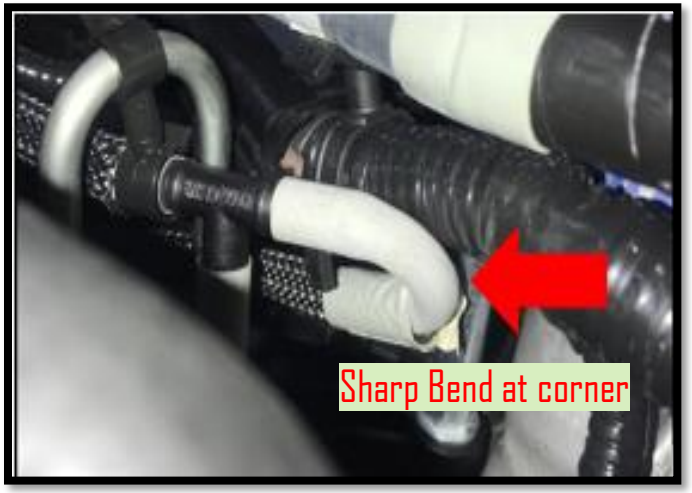
5.1 Identify Probable cause

Sr. no	Possible Causes
6	Diesel Overflow pipe Sharp Bend at corner
7	Distance clip sharp edge fouling to Diesel overflow Hose
9	Wiring pressure on Hose
10	Hose material not ok
13	Routine of hose/wiring not ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	1. Testing and Observations	Conclusion
6	Diesel Overflow pipe Sharp Bend at corner	Visual Checking	1) 20 vehicle check for hose routine check & monitoring. 2) After visual inspection in compartment sharp bend observed at corner . 3) Number of vehicles checked = 20 4) Number of vehicle found defective= 16 5) Number of vehicle found ok = 4	**Hypothesis Valid**



Observations of 20 vehicles checked

1) 20 vehicle check for hose routine check & monitoring.
2) After visual inspection in compartment sharp bend observed at corner .
3) Number of vehicles checked = 20
4) Number of vehicle found defective= 16
5) Number of vehicle found ok = 4

Inference:

With above observations it is concluded that after intended routine also hose bend at corner edge in all vehicle

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	1. Testing and Observations	Conclusion
7	Distance clip sharp edge fouling to Diesel overflow Hose	1. Visual Checking 2. Gap measure with Taper scale	1) Wiring distance clip to overflow pipe distance check with taper scale in 20 vehicles at FAI before engine top cover fitment . 2) Number of vehicles checked = 20 3) Number of vehicle found defective (0 mm fouling)= 15 4) Number of vehicle found ok = 05	**Hypothesis Valid**

CHECKSHEET

Desc : Wiring Distance clip to Diesel overflow Hose gap			Date : 12.12.2018	
Model : Marazzo				
Sr NO	CHECKPOINTS	WHAT TO CHECK	HOW TO CHECK	Observation \Result
1	Gap check	Wiring Distance clip to Diesel overflow Hose gap	Taper Scale	0 mm
2				0 mm
3				0 mm
4				0 mm
5				0 mm
6				5 mm
7				0 mm
8				2mm
9				0 mm
10				0 mm
11				4 mm
12				0 mm
13				0 mm
14				0 mm
15				3.5 mm
16				0 mm
17				0 mm
18				4 mm
19				0 mm
20				0 mm

Observations of 20 vehicles checked

1) 20 vehicle check for hose routine check & monitoring.
2) After visual inspection in compartment sharp bend observed at corner .
3) Number of vehicles checked = 20
4) Number of vehicle found defective (0 mm) = 15
5) Number of vehicle found ok = 5

Inference:

With above observations it is concluded that 80% cases wiring clip fouling to hose before Engine top cover fitment at FAI

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	1. Testing and Observations	Conclusion
9	Wiring pressure on Hose	1. Visual Checking 2. Touch & feel	1) Wiring distance clip to overflow pipe distance check with taper scale in 20 vehicles at FAI before engine top cover fitment & wiring pressure on diesel overflow pipe . 2) Number of vehicles checked = 20 3) Number of vehicle found defective= 15 4) Number of vehicle found ok = 05	**Hypothesis Valid**

Observations of 20 vehicles checked

1) Wiring distance clip to overflow pipe distance check with taper scale in 20 vehicles at FAI before engine top cover fitment & wiring .
2) Number of vehicles checked = 20
3) Number of vehicle found defective= 15
4) Number of vehicle found ok = 5

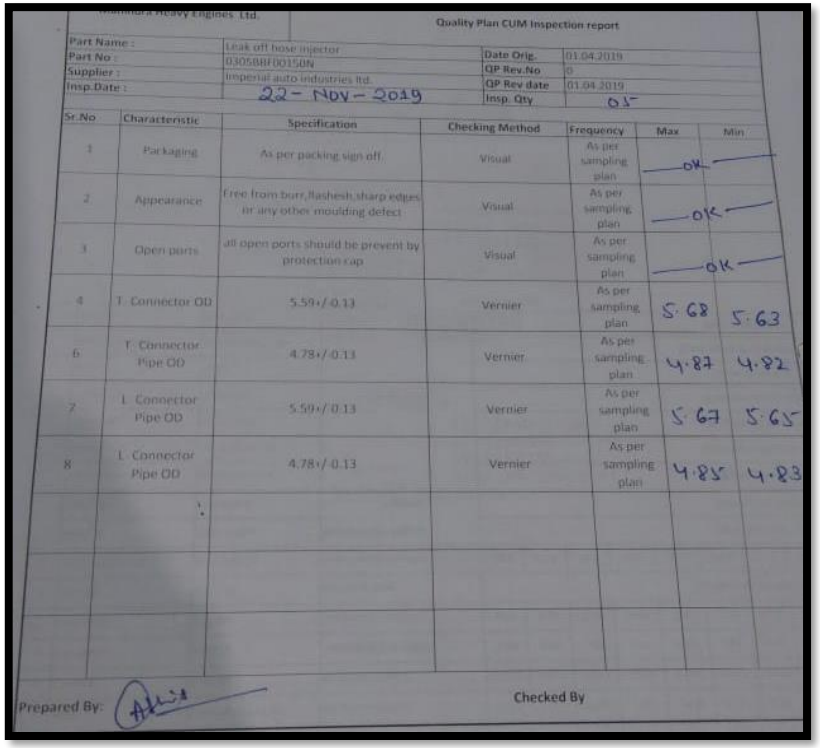
Inference:

With above observations it is concluded that 80% cases before Engine top cover fitment at FAI wiring pressure on Overflow pipe .

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	1. Testing and Observations	Conclusion
10	Hose not ok as per drawing	1. Visual Checking 2. Measurement	1) Parts check at supplier as per drawing	**Hypothesis invalid**



Mahindra Heavy Engines Ltd.

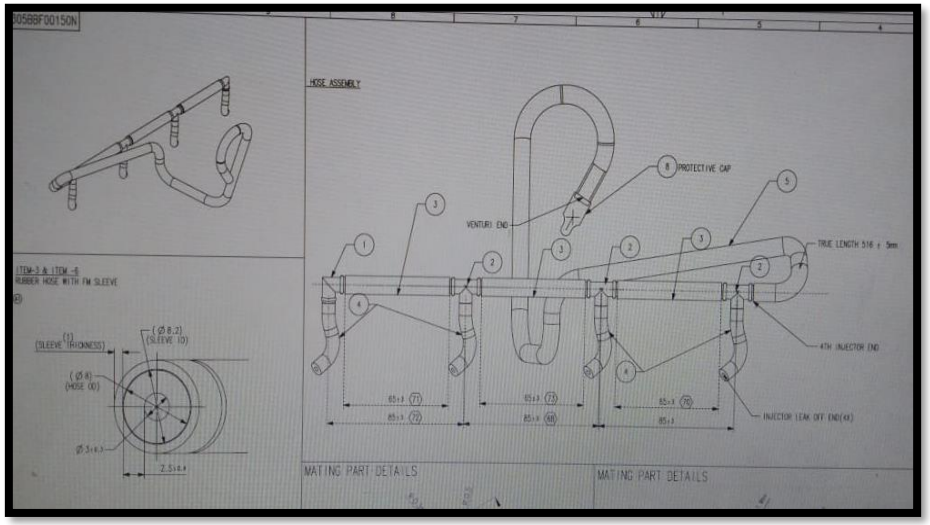
Quality Plan CUM Inspection report

Part Name :	Leak off hose injector	Date Orig.	01.04.2019
Part No :	03058BF00150N	QP Rev.No	0
Supplier :	Imperial auto industries ltd.	QP Rev date	01.04.2019
Insp.Date :	22-Nov-2019	Insp. Qty	05

Sr.No	Characteristic	Specification	Checking Method	Frequency	Max	Min
1	Packaging	As per packing sign off.	Visual	As per sampling plan	OK	
2	Appearance	Free from burr,flashesh,sharp edges or any other moulding defect	Visual	As per sampling plan	OK	
3	Open ports	all open ports should be prevent by protection cap	Visual	As per sampling plan	OK	
4	T. Connector OD	5.59+/-0.13	Vernier	As per sampling plan	5.68	5.63
6	T. Connector Pipe OD	4.78+/-0.13	Vernier	As per sampling plan	4.87	4.82
7	L. Connector Pipe OD	5.59+/-0.13	Vernier	As per sampling plan	5.67	5.65
8	L. Connector Pipe OD	4.78+/-0.13	Vernier	As per sampling plan	4.85	4.83

Prepared By: Checked By

Drawing



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

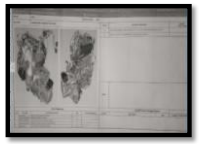
No	Probable Cause	Testing method	1. Testing and Observations	Conclusion
13	Routine of hose not ok.	1. Visual Checking	1) Routine of overflow pipe check as per SOS & design layout at chassis line after Engine preparation . 2) Number of vehicles checked = 20 3) Number of vehicle found defective= 00 4) Number of vehicle found ok = 20	**Hypothesis invalid**

CHECKSHEET

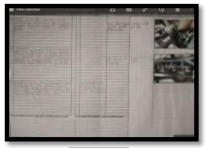
Desc : Diesel Oveflow Pipe routine as per SOS — Date : 12.12.2018

Model : Marazzo

Sr NO	CHECKPOINTS	WHAT TO CHECK	HOW TO CHECK	Observation \Result
1	Oveflow pipe routing	Overflow pipe routing as per SOS from below injector	visual	Ok
2				Ok
3				Ok
4				Ok
5				Ok
6				Ok
7				Ok
8				Ok
9				Ok
10				Ok
11				Ok
12				Ok
13				Ok
14				Ok
15				Ok
16				Ok
17				Ok
18				Ok
19				Ok
20				Ok



POS



SOS

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Cause No	Probable Cause	Testing method	Conclusion
1	Diesel Overflow pipe Sharp Bend at corner	1.Visual checking at Engine preparation stage chassis line	H[illegible]is valid
2	Distance clip sharp edge fouling to Diesel overflow Hose	1.Visual Chec[illegible] [illegible]easure with Taper [illegible]ale	Hypothesis valid
3	Wiring pr[illegible] [illegible]ose	1.Visual Checking 2.Touch & feel	Hypothesis valid
[illegible]	Hose material not ok	1. Part L1 analysis done	Hypothesis Invalid
5	Routine of hose not ok	1. Visual Checking	Hypothesis Invalid

Out of 5 causes 3 Valid Causes

WHY - WHY Analysis

5.3 Identify the root causes:

Cause No	Valid Probable Cause	WHY	WHY	WHY	Why	ACTION / COUNTERMEASURE
1	Diesel Overflow pipe Sharp Bend at corner	Diesel overflow pipe gets bend at protective sleeve end	Diesel overflow pipe protective sleeve length not sufficient to avoid sharp bend	Diesel overflow pipe length is as per CAD layout.	Design inadequate	Change the Design
2	Distance clip sharp edge fouling to Diesel overflow Hose	Clearance between clip edge & diesel overflow pipe is less	Clearance between clip edge & diesel overflow is per CAD layout.	Design inadequate		Clearance to be optimized
3	Wiring pressure on Hose	Wire rests on overflow pipe top surface	Wire routine is as per SOS	SOS inadequate		Wire routine to be redefine.

6. Implement the corrective measures

Possible Remedies Identified

↑ High **Legends** ↓ Low

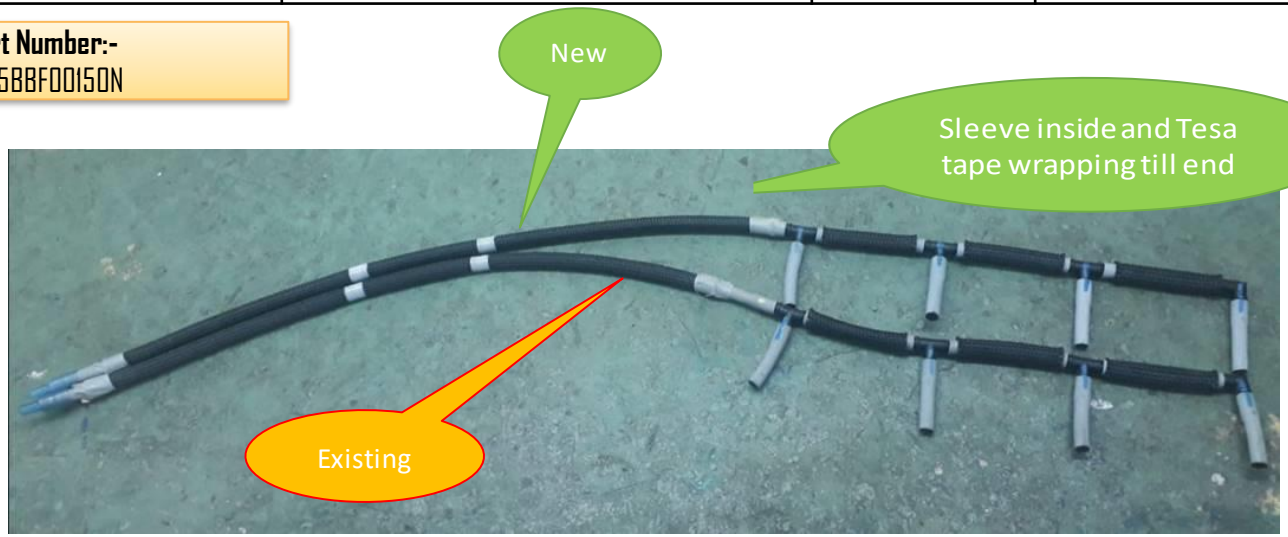
Sr No	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Design inadequate	Change the Design	High ↑	Low ↓	High ↑	Low ↓	Accepted
2	Design inadequate	Change the Design	Low ↓	Low ↓	Low ↓	Low	Accepted
3	SOS inadequate	Wire routine to be redefine.	Low ↓	Low ↓	Low ↓	Low ↓	Accepted

6. Implement the corrective measures

6.1 Actions Decided

Cause No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Design inadequate - Diesel Overflow pipe Sharp Bend at corner	Protective sleeve length 476 mm increase length to 500mm to cover till end	10.01.2019	SAWANT KAKASAHEB	CLOSED

Part Number:-
0305BBF00150N



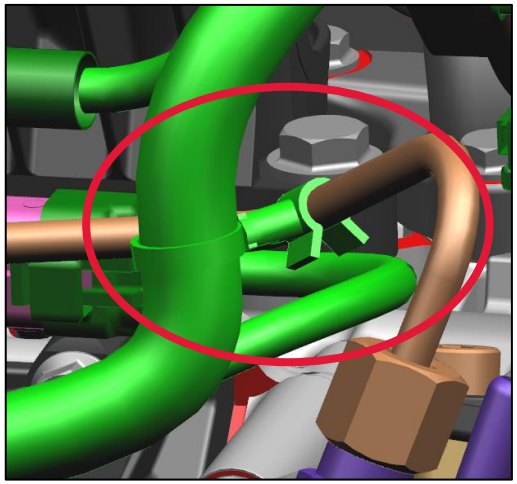
Before:-
Protective sleeve length 476 mm

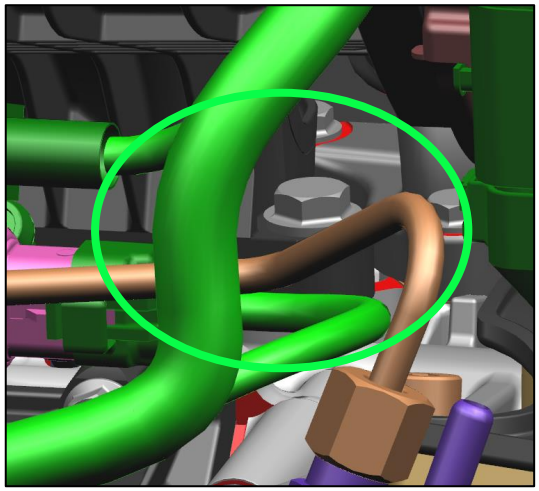
After:-
Protective sleeve length 500 mm

6. Implement the corrective measures

6.1 Actions Decided

Cause No	Root Cause	Action plan	Actual date	Responsibility	Status
2	Design inadequate - Distance clip sharp edge fouling to Diesel overflow Hose	1. Distance clip to be deleted to avoid the suspected cut of hose	5/12/2018	Dasoondi Pushpendra	CLOSED





Before:-
Engine Wiring distance clip fouling to Overflow pipe .

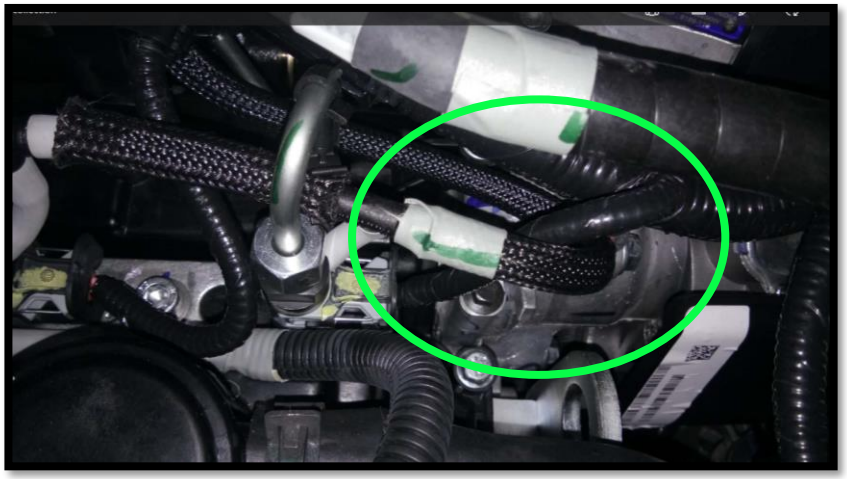
After:-
Engine Wiring distance clip deleted.

6. Implement the corrective measures

6.1 Actions Decided

Cause No	Root Cause	Action plan	Actual date	Responsibility	Status
3	SOS inadequate - Wiring pressure on Hose	The 4th injector wire to be routed below the overflow pipe at TCF engine preparation	5/12/2018	Suryabhan Sor	CLOSED

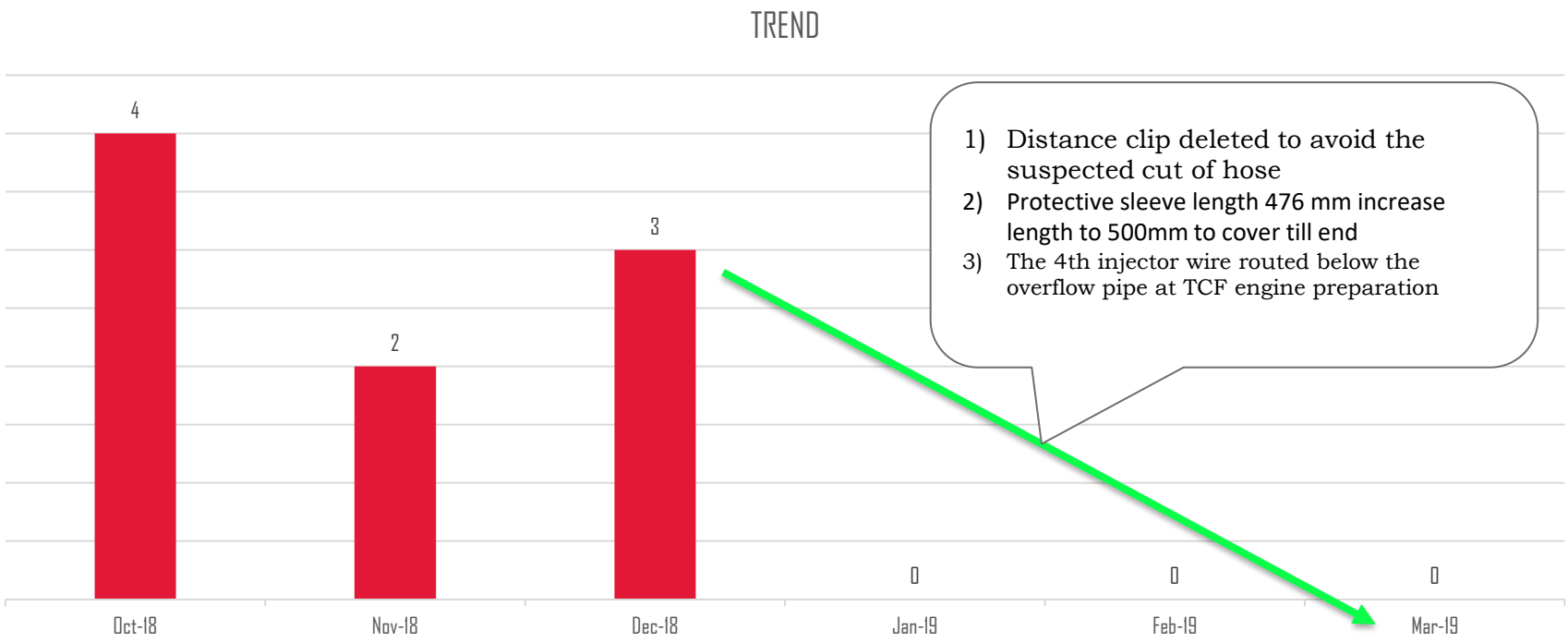
Before:-
Injector wire routing from above Overflow pipe



After:-
Injector wire routing from Below Overflow pipe.

7. Check & confirm the results & effects

7.1 Verify Results:



1) Distance clip deleted to avoid the suspected cut of hose
2) Protective sleeve length 476 mm increase length to 500mm to cover till end
3) The 4th injector wire routed below the overflow pipe at TCF engine preparation

Inference :- After Taking all Action Diesel leakage eliminated from Marazzo model

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

Sr No.	Action Taken	Target date	Responsibility	Status
1	Protective sleeve length 476 mm increase length to 500mm to cover till end	10.01.2019	SAWANT KAKASAHEB	Closed
2	Distance clip deleted to avoid the suspected cut of hose	05.12.2018	DASOONDI PUSHPENDRA	Closed
3	The 4th injector wire to be routed below the overflow pipe at TCF engine preparation	09.12.2018	Suryabhan Sor	Closed
4	PFMEA	09.12.2018	Suryabhan Sor	Closed
5	Control Plan	09.12.2018	Suryabhan Sor	Closed
6	QA Matrix	09.12.2018	Suryabhan Sor	Closed

Check & confirm the results & effects (Cont.....)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of Diesel leak customer complaint in Marazzo model.
2) 0 mis reduction.

In tangible Benefits :

1) Learning of CFT working approach
2) Gemba Culture is learned
3) Operator Morale Increased
4) Improved Customer Satisfaction.

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	Engineering change note	10.01.2019	Kaka Sawant	Closed
2	Engineering change note	05.12.2018	Dasoondi Pushpendra	Closed
3	Drawing	05.12.2018	Jayashankar	Closed
4	PFMEA updation	9.12.2018	Suryabhan Sor	Closed
5	SOS updation	9.12.2018	Suryabhan Sor	Closed
6	QA matrix Updation	9.12.2018	Suryabhan Sor	Closed

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Awareness given through OPL.	9.12.18	Suryabhan Sor	Closed

8.3 Additional Audit & check

1.	Check point added in buy off check sheet	14.12.2018	Rahul Gadekar	Closed

8.1 Updation of documents :- EN for sleeve length increased on diesel overflow pipe

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 1

Mahindra Rise.

Engineering Notice Release: E903-EN-MN02-000024
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date	26-Dec-2018
Initiated Date	27-Dec-2018

Engineering Notice No : E903-EN-MN02-000024

Released Date : 10-Jan-2019 11:50

Reference Change Req No : DCR042445

Location : MHEL-PUNE

Implementation Location : MHEL-PUNE

Engineering Notice Description : HOSE INJECTOR LEAK OFF

Reason : SLEEVE LENGTH INCREASED

Quantity : -(Regular)

Reason Code : Design robustness improvement

Priority : Mandatory

Note : IMMEDIATE

Affected Models : RDE903,RDE910,

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Released

Owner Name : M Krishnaprasad (23105523)

Engineer : M Krishnaprasad (23105523)

Approved By : 210304-V VIKRAMAN

Authorized By : E903-HEAD_301153-DAITHANKAR PARAG

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

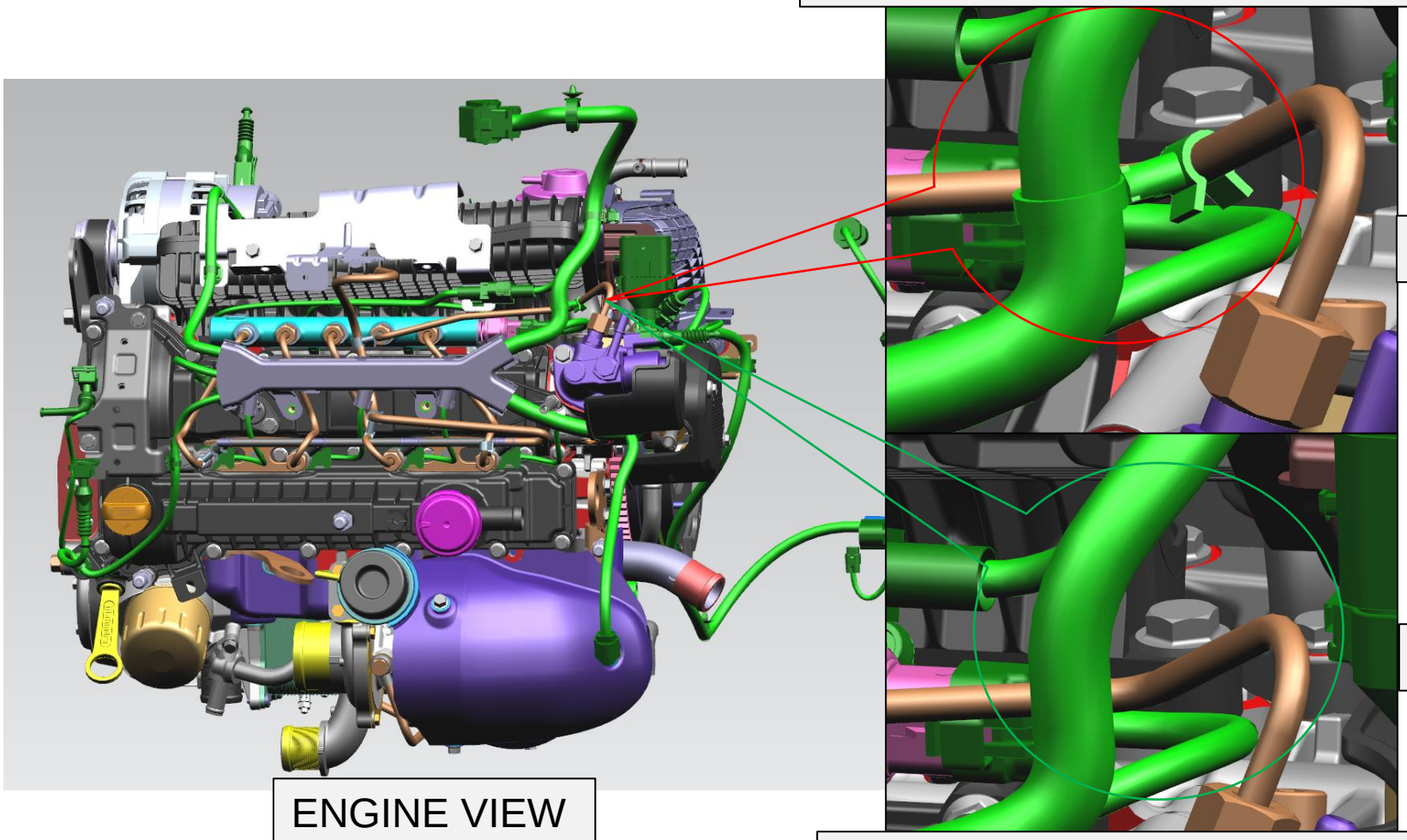
Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

Added/Deleted Qty : It's delta of existing released EN for the same parent part number

ICA :
O - New part number can be used in place of old part no. & vice versa;
I - New part number can be used in place of old part no. but not vice versa;
Z - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

8.1 Updation of documents: Distance clip deleted CAD layout

Distance clip before design change



BEFORE

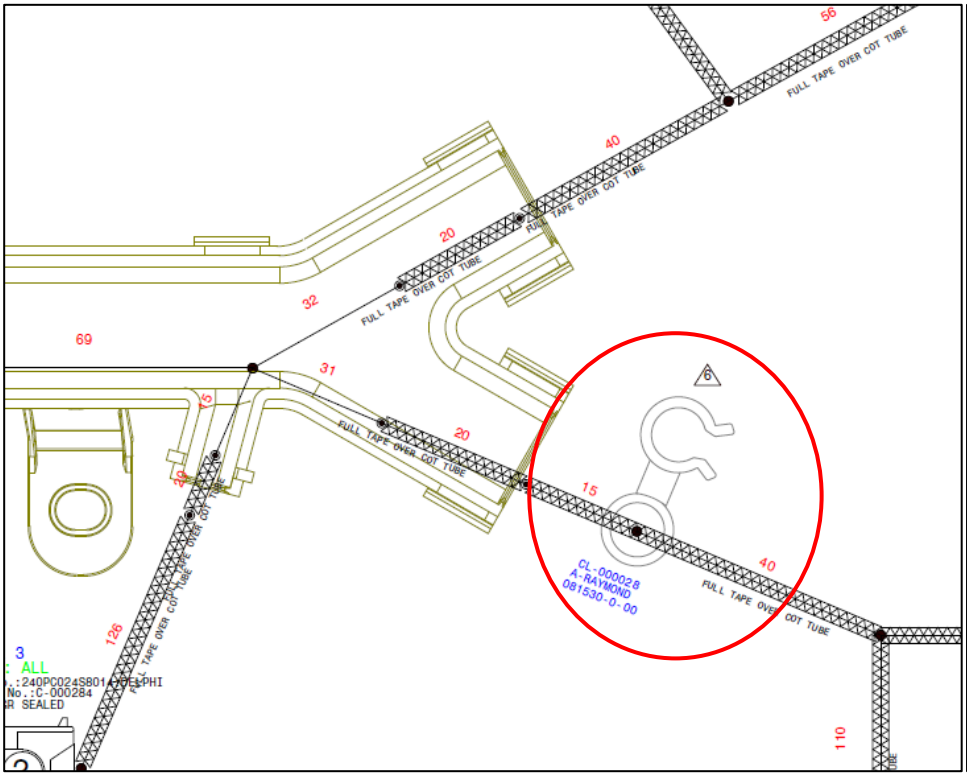
AFTER

ENGINE VIEW

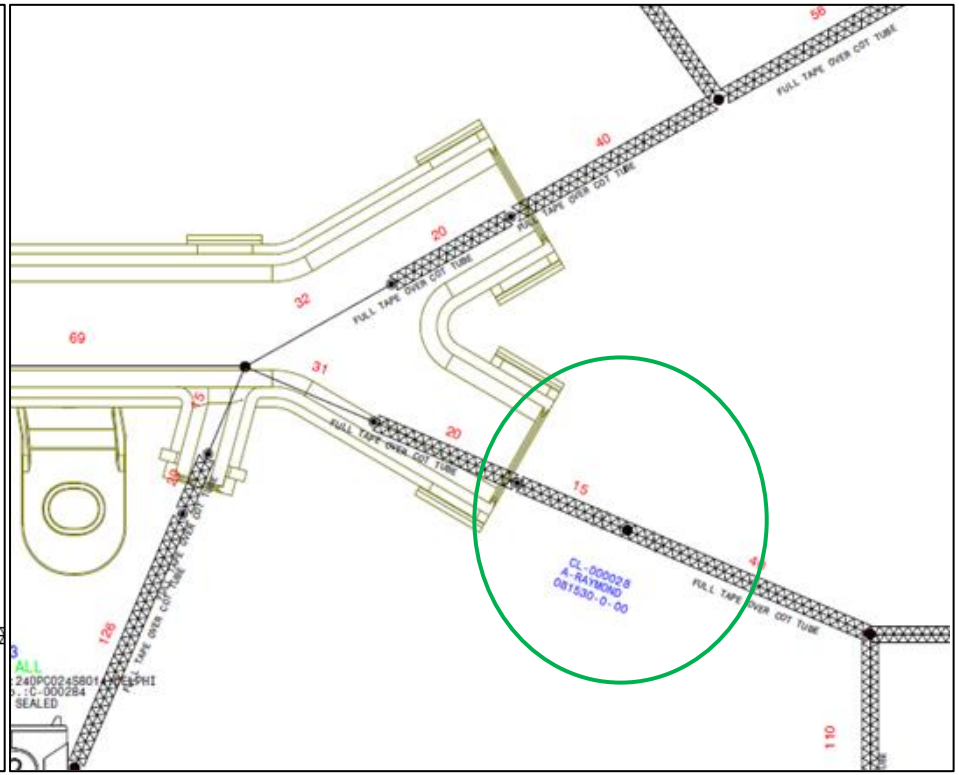
Distance clip after design change

8.1 Updation of documents: Drawing update – Distance Clip delete from wiring

BEFORE



AFTER



8.1 Updation of documents: PFMEA

Plant : **Nasik**			Model : **Marazzo**																	
Part No/s.:			Part Name/ Process: **Engine wiring**																	
Latest Change Letter: **NA**			Optn No.:																	
PU/ Shop : **Scorpio / Xylo**			Optn Desptn: **Engine wiring fitment on engine**																	
Module/ Cell : **TCF- Chassis**			**M/C Name & No.:**																	
Process Function	**Requirements**	**Potential Failure Mode**	**Potential Effect(s) of Failures**	**Sev**	**Class**	**Potential Cause(s)/Mechanism(s) of Failures**	**Current Process Controls Prevention**	**Occur**	**Current Process Controls Detection**	**Detect**	**RPN**	**Recommended Action(s)**	**Responsibility & Target Completion Date**	**Action Results**						
														Action Taken	**Sev**	**Occ**	**Det**	**RPN**		
Engine wiring fitment on engine	1)Tightening of positive cable nut.	1) Nut under torqued / loose.	**Mfg & Assy :** Engine does not start. **End User:** Sparking will happened	9	Z	1) Associate removes the tool before shutoff.	Only trained Associate deployed on stage.	2	Self Certification Buyoff Check	7	126									
				9		2) Improper power tool calibration	Calibration as per frequency decided in control plan.	2	Sticker for due date for calibration on tool.	7	126									
				9		3) Associate forgets to tighten nut on stud	Only trained Associate deployed on stage.	2	Self Certification Buyoff Check	7	126									
		2) Nut over slip	**Mfg & Assy :** Damaged to thread and rework	5		Nut not prefitted	Only trained Associate deployed on stage.	2	Self Certification Buyoff Check	7	70									
	4) Injector socket connections.	1) Injector Socket Connection poor lock or not done	**Mfg&Assy:** Vehicle No start or excess vibration and less pick up **End User** : NA	9	Z	Associate forget to lock the socket till click sound.	Only trained Associate deployed on stage.	2	Self Certification Buyoff Check	8	144									
		2) Injector socket wiring clipping on the injector pipe not done	**Mfg&Assy:** Post Assy Rework **End User :** Wiring Fouling to EGR Pipe	6		Associate does not lock the clip on the injector pipe . Poor locking.	Only trained Associate deployed on stage.	2	Self Certification Buyoff Check	7	84									
		3) injector mounting bracket bolt loose / not fitted	**Mfg&Assy:** Post Assy Rework **End User :** Engine cover fitment loose it comes under body noise	6		Associate forgets to tighten bolt or bolt not fitted	Only trained Associate deployed on stage.	2	Self Certification Buyoff Check	7	84									
		4) Dumpler not fitted / loose	**Mfg&Assy:** Post Assy Rework **End User :** Engine cover fitment loose so comes under body noise	6		Dumpler not fitted or loose	Only trained Associate deployed on stage.	2	Self Certification Buyoff Check	7	84									
		5) The 4th injector wire to be routed below the overflow pipe	**Mfg&Assy:** Diesel Pipe cut & leak **End User** : NA	9	Z	Associate forget to Route below overflow pipe	Only trained Associate deployed on stage.	2	Self Certification Buyoff Check	8	144									

Mahindra & Mahindra Ltd.

Process Potential Failure Mode & Effect Analysis.

Cross Functional Team

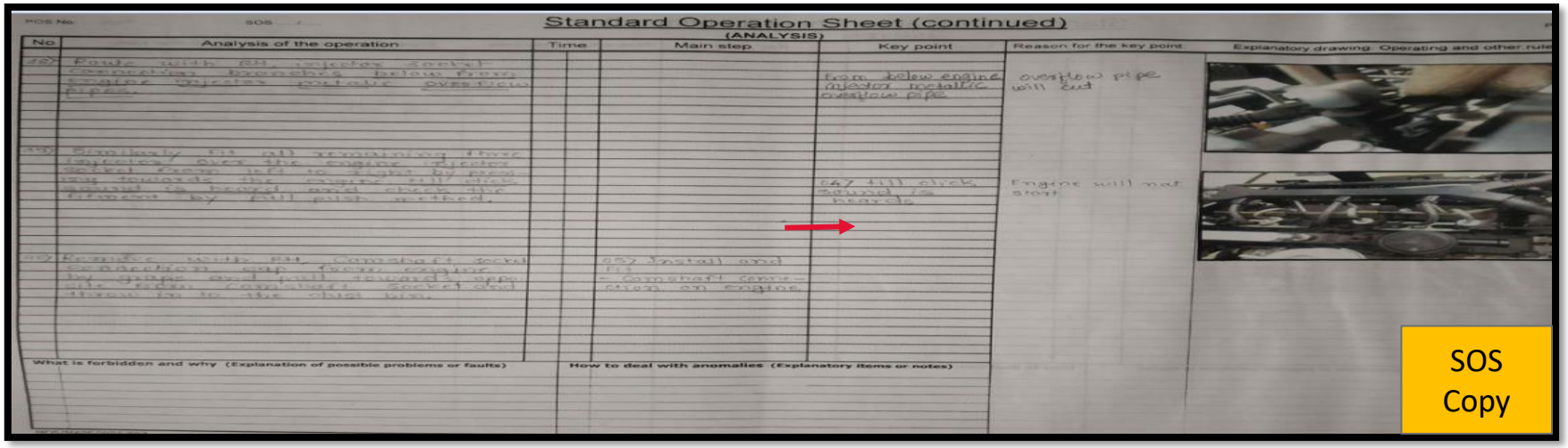
Nikhil / Suryabhan	**Hari Nagar**	
Vikrant Ghule	**Vikrant Morade**	

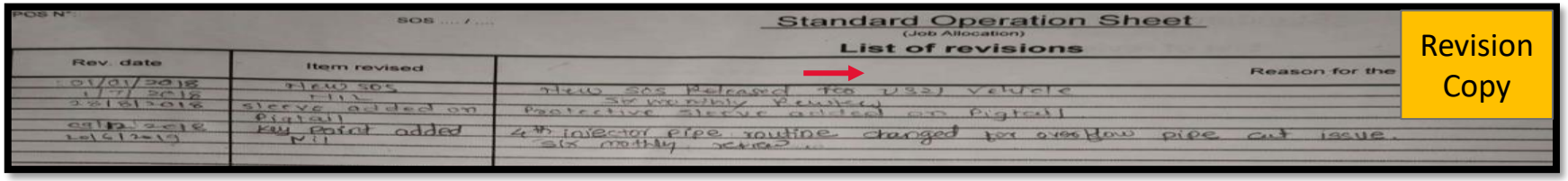
Reviewed by : **Nikhil Hatedkar**	Rev. No :	**4**	Doc No.:	
Approved by : **Taj Shaikh**	Rev. Date :	**5.8.19**	FMEA Date (Orig) : **25.05.2018**	

8. Standardization

8.1 Updation of documents: SOS

Sr No.	Action Plan	Target Dt	Resp.	Status
1	SOS revision done for injectors wire routine change.	09.12. 2018	Suryabhan Sor	Closed





8. Standardization

8.1 Updation of documents: QA MATRIX

QA MATRIX

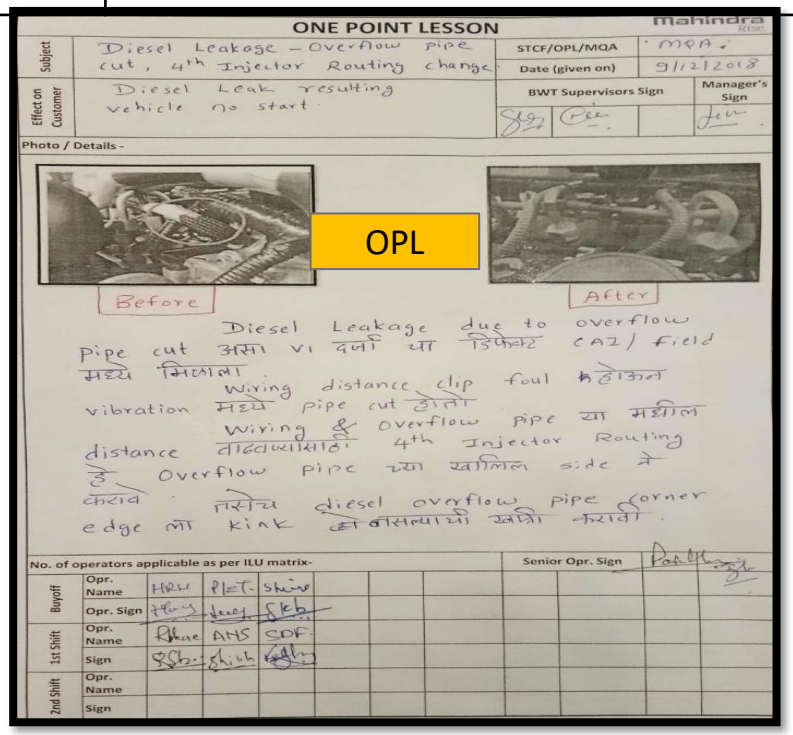
PU NAME	SCORPIO	INITIALIZATION DATE	Global Quality Guarantee level.
DEPT NAME	SXTCF	SIGNATURES	
SECTION NAME	CHASSIS	Dept Head: Rahul narkar	Section: 100
		Module Manager: TAJ SHAIKH	Plant: 100
		Section Manager: 2 SP	Revision date:

Workstation No.	Process description (POS)	What To Check	Date	Failure modes	Severity	Prev. occurence	Potential defect	Inspection process – Section: CSC FITMENT	CLUTCH & PRESSUI	STARTOR MOTOR	TC HOSE	ENGINE WIRING	INTERCOOLER	BUY OFF	Outside section: FAI	CAI				Guarantee value: Workstation	Section	Outside section	PLANT	Faults recorded: Section (1M)	Outside section (1M)	CUSTOMER (3M)	Quality guarantee level: WORKSTATION	Section	Outside section	PLANT	Comments
	Diesel Leak	The 4th injector wire to be routed below the overflow pipe at TCF engine preparation	09.12.2019	Fouling	5							⊙ 3	⊙ 3		⊙ 3	▯ 1				3	6	4	10				NOK	OK	NOK	Ok	

8. Standardization

8.2 Training:

Sr No.	Action Plan	Target Dt	Resp.	Status
1	1.OPL given to Engine preparation Stage Associate.	09.12.2018	Suryabhan Sor	Closed



8. Standardization

8.3 Additional Audit & check

SN	Document	Actual date	Responsibility	Status
2	Check point added at FAI before engine top cover fitment, Under hood checking	14.12.18	Rahul Gadekar	Closed

MARAZZO CONSOLE STAGE – MARKING STANDARD

		Description	What to Check		How to Check	Why To check	How to Mark	
2	SC	Positive cable mounting on Battery.	1.Correct orientation of Positive cable. 2.No loose fitment of Positive cable. 3.Positive cap fully lock.	V1	F	1.Will foul to other parts. 2.Vehicle will not start. 3.Water entry in fuse box		Relative Marking
3		Battry mounted fuse box.	1. No Loose fitment. 2.No loose fitment of 80 AMP fuse wire.(2 Nos.) 3.Fuse box cover fully lock.	V1+	1.Pull Push and marking. 2.Pull Push and marking. 3.Touch and feel	1.Get damage 2.Vehicle Inoperateable. 3.Water entry in fuse box		Relative Marking
6		Engine and Engine room wire loom interconnection.	1.Completly lock. 2.Completly mounted on engine bracket.	V1	1.Pull Push and marking. 2.Visual	1.Vehicle will not start. 2.Socket get damge over the period		Relative Marking
12		Heater Hose Connection	1.As per position (Upper/Lower) 2.Hose completely lock on HVAC	V1	1.Visual 2.Pull push and Marking	1.Vehicle colling will affect 2.Coolant leak		Relative Marking
13		Brake booster hose dislocated	Booster hose routing through clamp	v1	Visual marking	Field		Relative marking
14		Fuel return line	No cut mark on fuel line	V1	1.Visual 2 feel and Marking	1.Diesel leak 2. P126 defect		Relative Marking
15		Radiator hose filmed as engine	Coolnt inserted Hoseke marking matched	V1	Visual ,Marking	Field		Relative Mark

Page 1

9. Summarize & Plan future actions

9.1 Conclusion:

Diesel leak concern eliminated with correction in

1. Protective sleeve length increase
2. Engine wiring clip delete

9.2 Horizontal Deployment:

Horizontal deployment done at S-201

9.3 Lessons Learnt:

Things Gone Wrong :

- While designing diesel overflow pipe kink at corner was not anticipated
- While designing Engine wiring routing distance clip fouling not consider which causing cut in running due to vibration.

Things Gone Right :

- After analysis & brainstorming the sleeve length modification implementation eliminated the Overflow pipe kink at corner.
- The Distance clip deletion from engine wiring eliminated the fouling & pipe cut concern in vibration

Thank you

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